

云端服务费用报告

Cloud Service Cost Report

2026 年 4 月 – 7 月

April – July 2026

提交对象: mmrunners.org 理事会
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一、核心结论

Bottom Line

我们的照片系统跑在 Google 云上。到目前为止，四个月总共大约花了 11 到 27 美元。我们设定的上限是每月 20 美元，从来没有超过。

Our photo system runs on Google Cloud. Over four months it has cost roughly US\$11–27 in total. Our cap is US\$20 per month, and we have never exceeded it.

六月比七月贵。六月有两个设置上的失误，都已经找到并修好了。七月的数据可以证明修复有效。June cost more than July. Two setup mistakes happened in June. Both were found and fixed, and the July figures prove the fixes worked.

真正需要留意的不是钱，而是免费额度。Google 每月给我们一定的免费用量，用完之后才开始收费。我们的费用之所以这么低，是因为几乎所有用量都刚好在免费额度以内。七月我们用掉了免费计算额度的 90%，已经很接近上限。

The thing to watch is not the money — it is the free monthly allowance. Google gives us a set amount of free usage each month, and we only pay once it runs out. Our bill is low because nearly all our usage fits inside that allowance. In July we used 90% of the free computing allowance, which is close to the limit.

二、账单：六月与七月

The Bill: June and July

月份 Month	估算费用 Estimated cost	说明 What drove it
4 月 April	\$0	系统尚未上线 Not live yet — nothing was running
5 月 May	\$0	系统尚未上线 Not live yet — nothing was running
6 月 June	\$8 – \$15	两个设置失误（见第三节） Two setup mistakes (see Section 3)
7 月 July (截至 28 日 to 28th)	\$3 – \$12	失误已修复；费用主要是照片存储 Mistakes fixed; mostly photo storage

月份 Month	估算费用 Estimated cost	说明 What drove it
合计 Total	\$11 – \$27	预算上限：每月 \$20 Budget cap: \$20 per month

重要说明：以上是估算值，不是 Google 的正式账单金额。 Important: these are estimates, not Google's invoiced amounts.

为什么只能估算？因为在这几个月里，我们没有开启 Google 的账单明细导出功能，所以详细的花费记录只存在于 Google 的网页后台，无法直接取出。我们改用「实际用量 × 公开价目表」的方式推算，用量数字是准确的，价格是公开的，但两者相乘后仍有误差空间。

Why only estimates? During these months we had not switched on Google's billing detail export, so the itemised record exists only inside Google's web console and cannot be pulled out directly. We calculated instead from actual usage multiplied by Google's published price list. The usage figures are exact and the prices are public, but the result is still an approximation.

主要误差来自网络流量。我们能看到总流量，但看不出其中哪些是免费的（在 Google 内部搬运数据）、哪些是收费的（传送到外部网络）。这部分的不确定性大约是每月 0 到 10 美元，也是上表金额呈区间的原因。第五节说明如何彻底解决这个问题。

The main uncertainty is network traffic. We can see the total, but not which part was free (data moving inside Google) versus billed (data sent out to the public internet). That gap is worth roughly \$0–10 per month, which is why the figures above are ranges. Section 5 explains how to close it permanently.

三、六月发生了什么

What Happened in June

问题一：搜索服务器被设为全天候开机 Problem 1: The search server was left switched on around the clock

当跑者上传自拍来找自己的照片时，会用到一台「人脸搜索服务器」。为了让第一次搜索快一点，这台服务器被设定成永不关机。结果是：不管有没有人在搜索，它 24 小时都在计费。

When a runner uploads a selfie to find their photos, a face-search server does the work. To make the first search faster, it had been set never to switch off. The result: it billed us 24 hours a day whether anyone was searching or not.

这个状况持续了三天半（6 月 17 日至 20 日），花费约 4.46 美元——是六月最大的一笔。如果一直没发现，大约是每天 1.70 美元、一年 50 美元，而且这段时间几乎没有人在使用它。

This ran for three and a half days (17–20 June) and cost about US\$4.46 — the single largest item in June. Left in place it would have cost roughly \$1.70 a day, about \$50 a year, for a server that was almost entirely idle.

如何修复：关掉「永不关机」的设定，改成没人使用时自动关闭。我们也定下一条规则：任何在闲置时还在花钱的服务，都不允许存在。代价是每次久未使用后的第一次搜索会稍慢几秒，这个取舍我们接受。

The fix: we removed the never-switch-off setting so the server now shuts down when idle, and adopted a rule that nothing may cost money while sitting idle. The trade-off is that the first search after a quiet period takes a few seconds longer. We accept that.

问题二：照片走了一条昂贵的线路 Problem 2: Photos travelled by an expensive route

当跑者点「保存到相册」或看原始大图时，照片原本会先经过我们的服务器再送到手机。这条路线让同一份照片被收费两次——出我们服务器算一次，再送给用户算一次。

When a runner tapped "Save to Photos" or opened a full-size image, the photo used to pass through our own server before reaching their phone. On that route the same photo was billed twice — once leaving our server, and again on delivery.

6 月 22 日有一场比赛活动，那一天就传了 22 GB，约合 2.54 美元。这个问题很容易被忽略：如果看的是全年费用曲线，这一天会被平均掉，几乎看不出来。而且花费会随着跑者下载的照片数量增加，活动越多、越贵。

On 22 June there was a live event, and 22 GB moved in that single day — about US\$2.54. This was easy to miss: on an annual cost chart, one expensive day averages away to almost nothing. And the cost grows with the number of photos runners download, so a busy season would multiply it.

如何修复：照片现在由储存空间直接传给跑者的手机，不再经过我们的服务器。批量下载也改由浏览器自行打包。跑者的使用体验完全没有改变。

The fix: photos now go straight from storage to the runner's phone, bypassing our server entirely, and bulk downloads are assembled by the browser itself. The experience for runners is unchanged.

修复效果（有数据佐证） Proof that the fixes worked

项目 Measure	6 月 June	7 月 July	结果 Result
照片传输量 Photo traffic on the costly route	25.1 GB	0.6 GB	下降 97% Down 97%

项目 Measure	6 月 June	7 月 July	结果 Result
服务器闲置计费 Idle server billing	很高 High	接近零 Near zero	已消除 Eliminated

七月的活动量比六月还大，但费用反而更低，因为这两项修复都生效了。
July was busier than June, yet cost less — because both fixes were working.

四、七月还发生了一件事

One More Thing That Happened in July

7 月 27 日至 28 日，我们遇到一次照片遗失事件。这主要是可靠性问题，不是费用问题，但理事会应该知道，因为它用掉了七月大部分的免费额度。
On 27–28 July we had a photo-loss incident. It was mainly a reliability problem rather than a cost problem, but the board should know about it because it consumed most of July's free allowance.

发生了什么：我们的自动部署流程把一项传输时间上限错误地设成了 60 秒，而正确值应该是 30 分钟。结果有一半的照片传输在中途被强制中断，1,188 张志工上传的照片（约 5.1 GB）卡在暂存区，没有进到相册，因此在网站上看不到。
What happened: our automated deployment set a transfer time limit to 60 seconds when it should have been 30 minutes. Half of all photo transfers were cut off part-way, leaving 1,188 volunteer-uploaded photos (about 5.1 GB) stranded in temporary storage — never filed into the gallery, and therefore invisible on the site.

结果：1,188 张照片全部找回并恢复。有 9 张永久遗失 —— 系统误判它们已经安全备份，因而删除了暂存的副本。
Outcome: all 1,188 photos were recovered. Nine were permanently lost — the system wrongly concluded they were already safely stored and deleted the temporary copies.

已修复：传输时间上限已改回 30 分钟，而且部署流程现在会主动检查这个数值，不符就中止上线。更重要的是，除非确认照片确实已经安全存好，系统绝不删除暂存副本 —— 多留一份备份几乎不花钱，删错一张就再也找不回来。
Fixed: the time limit is back to 30 minutes, and the deployment process now checks this value and refuses to go live if it is wrong. More importantly, the system will never delete a temporary copy unless it has confirmed the photo is genuinely stored safely — keeping a spare copy costs almost nothing, while deleting the only copy is permanent.

费用影响：这次事件本身花费很少，但为了恢复照片而重新整理索引，用掉了七月免费计算额度的 90%。如果同一个月再来一次类似事件，我们就会真的产生费用。

Cost impact: the incident itself cost little, but re-processing the photos to recover them consumed 90% of July's free computing allowance. A second such event in the same month would have produced a real bill.

五、未来预期

What to Expect Going Forward

情况 Situation	预期月费 Expected monthly cost	说明 Notes
没有比赛活动的月份 A quiet month, no events	\$2 – \$4	主要是照片存储 Mostly photo storage
有比赛活动的月份 A month with live events	\$4 – \$8	跑者下载照片会增加流量 Runner downloads add traffic
清理旧软件副本之后 After old software cleanup	每月省约 \$0.70 Saves about \$0.70/month	已准备好，待启用 Ready, awaiting activation
预算上限 Budget cap	\$20	达到 50%、85%、100% 时会发邮件提醒 Email alerts at 50%, 85%, 100%

随着活动场次增加，费用会跟着上升，但增幅不会是平顺的。因为免费额度的关系，费用是「阶梯式」跳动的：只要用量还在额度内就几乎免费，一旦超出就会明显跳一级。所以我们要盯的是「还剩多少免费额度」，而不是「这个月花了几块钱」。

Costs will rise as we run more events, but not smoothly. Because of the free allowance, cost moves in steps: while usage stays inside the allowance it is nearly free, and once it goes over, the bill jumps noticeably. So what we monitor is how much free allowance is left, not how many dollars we have spent.

六、需要改进的事项

What We Are Changing

事项 Action	为什么重要 Why it matters	状态 Status
开启账单明细导出 Switch on billing detail export	这是本次检视中最不该发生的一件事：我们无法直接回答「六月花了多少」，必须靠推算。 开启后每一笔花费都有据可查。 The most uncomfortable finding of this review: we could not answer "what did June cost" without reconstructing it. Once on, every charge is auditable.	待理事会指定人员点击开启（网页后台操作） Needs one click in the web console
导出 4–7 月正式账单 Export the April–July invoices	用 Google 的正式数字取代本报告中的所有估算值。 Replaces every estimate in this report with Google's audited figures.	待办 To do
启用旧软件副本自动清理 Switch on automatic cleanup of old software copies	每次更新都会留下一份旧副本，从未清理，已累积 111 份、12.4 GB，每月约 \$1.28 且持续增长，属于纯浪费。 Every update left an old copy behind and none were ever removed — 111 copies, 12.4 GB, about \$1.28/month and rising. Pure waste.	已设定并完成安全验证，试运行中，待正式启用 Configured and safety-checked; in trial mode, awaiting activation
建立每日与每周检查 Set up daily and weekly checks	六月那次单日流量高峰，在年度曲线上几乎看不见。只有按日检查才能当天发现问题。 June's one-day traffic spike was nearly invisible on an annual chart. Only a daily check catches that on the day.	清单已完成，待排入日常 Checklist written; to be scheduled
修正三处「文件写的与实际不符」的设置 Correct three settings that were documented wrongly	这正是照片遗失事件的根本原因：文件写的是 30 分钟，实际跑的是 60 秒，两边都被当成正确。同类问题另外还有三处。 This is exactly what caused the photo loss: the document said 30 minutes, production ran 60 seconds, and both were believed. Three more of the same kind were found.	待办 To do

七、一句话总结

In One Sentence

费用本身很低而且在控制之内；真正的问题是我们过去看不清自己花了什么钱，而两次事件都源自「文件写对了、实际设错了」。这两个缺口现在都有明确的补救措施。

The cost is low and under control; the real problem was that we could not see what we were spending, and both incidents came from settings that were documented correctly but deployed incorrectly. Both gaps now have concrete fixes in hand.